

PRODUCT REQUIREMENT DOCUMENT (PRD)

Autonomous Multi-Agent Social Media Agency Pipeline

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1. Project Overview & Strategy Blueprint

The objective of this project is to architect, develop, and deploy an automated, fully collaborative multi-agent AI framework simulating a full-cycle creative social media marketing agency. The core platform accepts unstructured client input, performs rigorous competitive landscape analysis, develops a comprehensive monthly calendar strategy, translates the timeline into structured weekly assignments, generates professional psychological-hook copy, automates creative visual assets generation, and manages an integrated human-in-the-loop validation gate prior to final multi-platform publishing.

The Strategic Core: Anti-Hallucination Engineering Framework

To prevent downstream structural breakdown and compounding errors commonly found in traditional multi-agent chains, this PRD enforces an isolated, validated data layer between every individual hand-off step.

This strict structural framework utilizes four main mitigation pillars:

- **Schema Contracts (Pydantic Gating):** Agents are prohibited from communicating via raw text streams. Every handoff must conform strictly to a validated JSON architecture.
- **Adversarial Quality Control:** Dedicated validator agents evaluate operational outputs independently against client parameters before downstream processing is unlocked.
- **Evidence Tracking:** Every claim, trend analysis, or strategy choice made by an analytical agent must reference active URLs or verified context logs.
- **Deterministic Human Verification Gate (HITL):** Hard software limitation preventing any publishing action until an explicit manual state change is made inside the content database.

2. System Flow & Data Conveyor Belt

The operational sequence transforms chaotic input parameters into clean, structured digital assets via a synchronized architectural pipeline:



3. Comprehensive Agent Technical Specifications

Agent 1: The Account Manager (AM)

- **Objective:** Standardize initial unstructured briefing details into a centralized Brand Dossier schema object.
- **Guardrail Mechanism:** Enforces absolute required property checks. If essential brand components are missing, the AM outputs an immediate **INSUFFICIENT_CONTEXT** state flag and halts execution rather than guessing parameters.

Agent 2: The Strategist

- **Objective:** Conduct targeted live crawling operations, identify concurrent competitor patterns, define exactly 3–5 operational Content Pillars, and generate a macro monthly distribution plan.
- **Guardrail Mechanism:** Strict URL and evidence citation requirement. Every pillar item requires a structural justification mapping text directly back to the verified data context.

Agent 3: The Strategy Validator (Adversarial Gate)

- **Objective:** Perform cross-examination of the Strategist's data outputs against original client boundaries.
- **Guardrail Mechanism:** Acting as a programmatic firewall, this agent executes validation assertions. If a conflict or missing citation occurs, it issues an execution rejection and routes the context backward with structured correction feedback notes.

Agent 4: The Project Manager (PM)

- **Objective:** Parse monthly macro structural blueprints into localized, weekly workflow data modules.

Agent 5: The Copywriter

- **Objective:** Formulate target text parameters, organic hook variations, strategic platform hashtag blocks, and explicit literal visual descriptions.
- **Guardrail Mechanism:** Prompt constraints isolate the system environment from fabricating external operational claims. Copywriters are technically constrained to use information strictly inside the verified database.

Agent 6: The Designer Engine

- **Objective:** Read visual copywriter descriptions and translate parameters into targeted visual image API generation pipelines.

4. Schema Contracts & Data Validation Layer

To eliminate programmatic drift across the pipeline runtime, all communication interfaces must execute via structured Pydantic models. Data is strictly managed through the following definitions:

```
from pydantic import BaseModel, HttpUrl, Field
from typing import List, Optional

class ContentPillar(BaseModel):
    pillar_name: str = Field(description="Name of the core brand content category")
    justification: str = Field(description="Direct logical connection to the verified client data profile")

class PostStructure(BaseModel):
    date: str
    platform: List[str]
    pillar: str
    angle_and_objective: str

class StrategySchema(BaseModel):
    client_name: str
    competitor_urls_scanned: List[HttpUrl]
    market_trends_discovered: List[str]
    content_pillars: List[ContentPillar]
    one_month_calendar_skeleton: List[PostStructure]

class CopyAndDesignSchema(BaseModel):
    post_id: str
    scheduled_date: str
    target_platforms: List[str]
    hook_text: str
    body_caption: str
    hashtags: List[str]
    visual_generation_prompt: str = Field(description="Literal description for generation engine. Abstract text/metaphors are illegal.")
```

5. Post-Production, Approval Layer, & Publishing Infrastructure

The Content Conveyor Belt (Airtable Integration)

Upon automated generation of text copy elements and corresponding visual asset link locations, the system compiles the aggregate structural payload into a dedicated row entry inside an operational Airtable database.

- The record initializes under a default system state: "Pending Human Review".
- An automated webhook engine broadcasts a workspace alert (via Slack, WhatsApp, or email) to notify you of incoming material needing manual configuration or validation checks.

The Execution Trigger Pipeline

1. The administrator analyzes visual performance qualities and text formats on the main workspace control panel dashboard.
2. Any necessary manual adjustments are applied dynamically to cell objects inside the target database row.
3. Upon changing the status flag to "Approved", an automated Make.com listener script intercepts the status update.
4. The script immediately maps parameters into social scheduling suite APIs (e.g., Buffer, Hootsuite, Publer) to handle seamless queuing at the defined publication timestamp.

6. Risk Management & Automated Error Resolution

Risk Scenario / System Event	Automated Resolution Mechanism
Competitor Data Hallucination	Strategy Validator intercepts missing or unverified URLs, immediately dropping the pipeline segment back to the search layer under strict parameter validation modes.
Image Generation Text Corruptions	Copywriter system engines strip typographic rendering demands out of visual prompts. Text layout properties are handled downstream via programmatic HTML overlay canvas rendering steps or predefined design templates.
API Workspace Timeout Limits	Automated architecture switches runtime contexts dynamically, routing execution tasks from high-end primary language frameworks to isolated fallback engines to maintain loop integrity.

7. Technical Stack Matrix Summary

- **Pipeline Orchestration Engine:** CrewAI + CrewAI Flows (enforcing state logic loops).
- **Primary Linguistic Reasoning Engine:** Anthropic Claude 3.5 Sonnet (for advanced strategy parsing, compliance assertions, and copywriting optimization).
- **Visual Generation Middleware:** Midjourney / Flux API (for customized generative brand artwork) or Bannerbear (for structured rule-based template management).

- **Database and Control Workspace:** Airtable REST API.
- **Integration Engine:** Make.com webhooks.